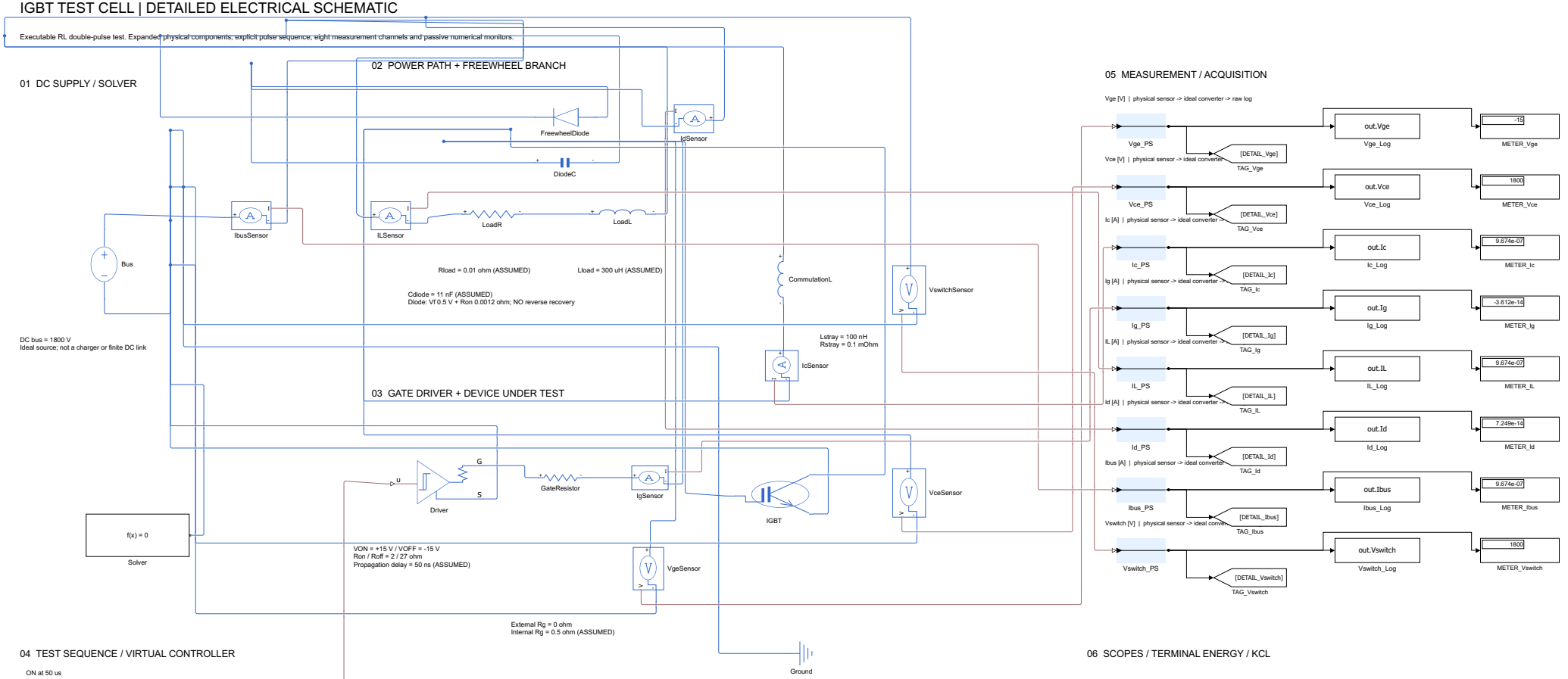
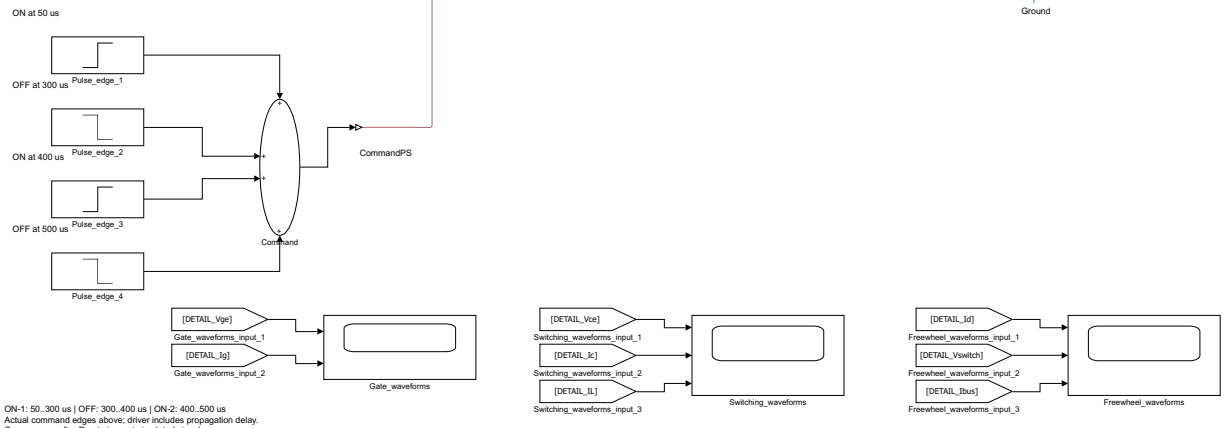
IGBT TEST CELL | DETAILED ELECTRICAL SCHEMATIC

Executable RL double-pulse test. Expanded physical components, explicit pulse sequence, eight measurement channels and passive numerical monitors.

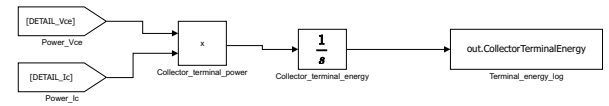


04 TEST SEQUENCE / VIRTUAL CONTROLLER

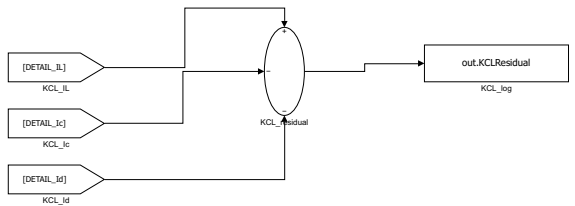


ON-1: 50..300 us | OFF: 300..400 us | ON-2: 400..500 us
Actual command edges above; driver includes propagation delay.
Open scopes after Run to inspect simulated signals.

06 SCOPES / TERMINAL ENERGY / KCL



Integral of Vce*Ic is collector TERMINAL energy.
It includes conduction and reactive transfer; not calibrated pure Eon/Eoff.



KCL = IL - Ic - Id. Ideal sensor model; no ADC quantisation or noise.
Auxiliary AC/DC, 24 V supply, optical isolation, CAN/USB and interlocks
are not represented as hardware in this fixture. Hardware decision: INCONCLUSIVE.

Representative CM1500HC-66XB; NOT identified Wabtec module
Equation-based full I-V and charge model; fixed 25 degC
Vth input 6.3 V; calibration Vce 2.6 V at 1500 A / 25 degC
Cies / Cres / Coes = 164 / 1.4 / 11 nF
Transit time 0.2 us (ASSUMED); dynamic calibration remains incomplete